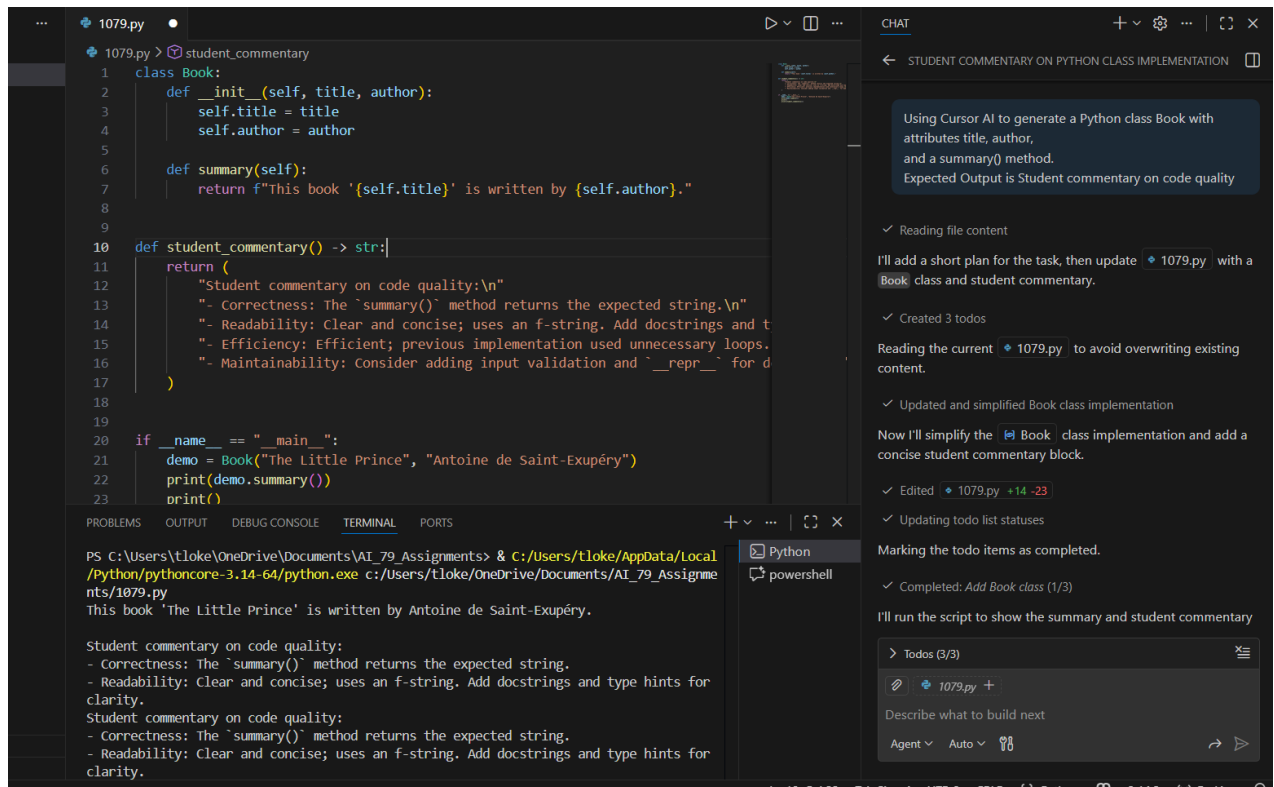
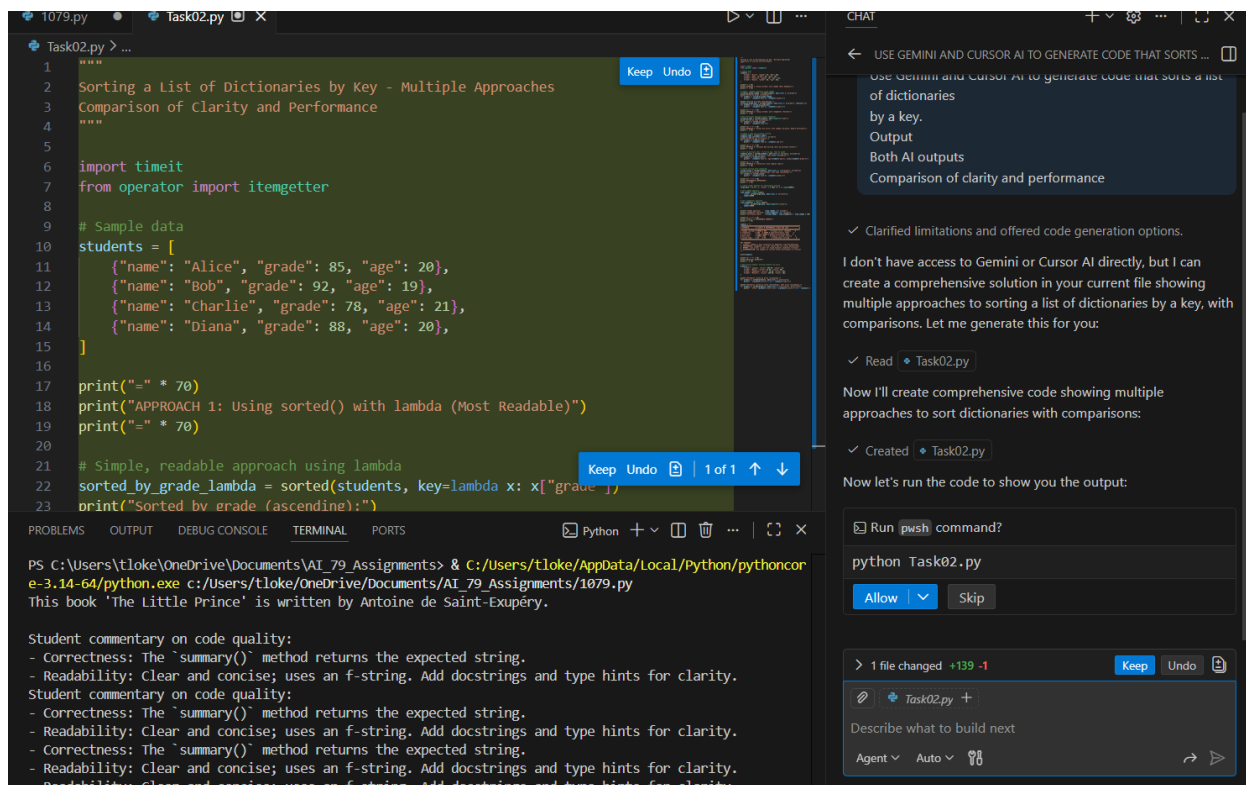


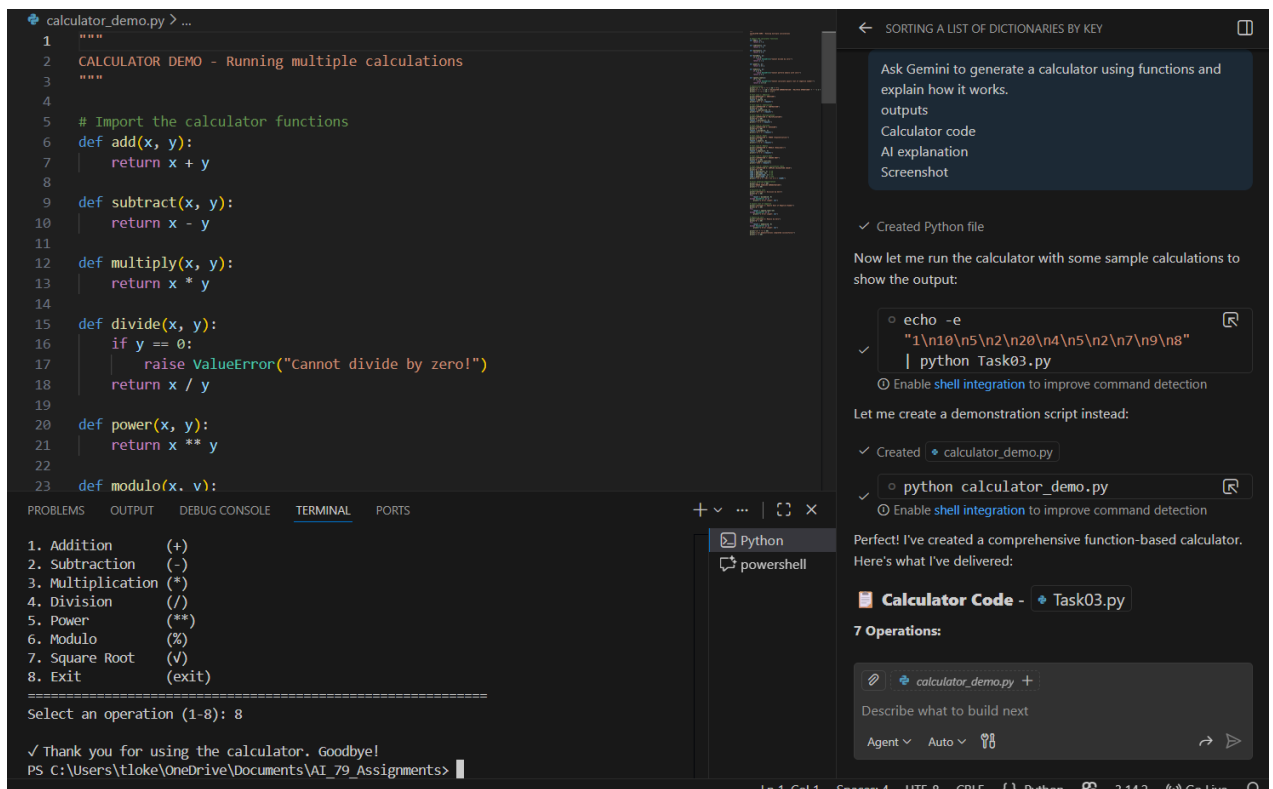
Task 1: Book Class Generation



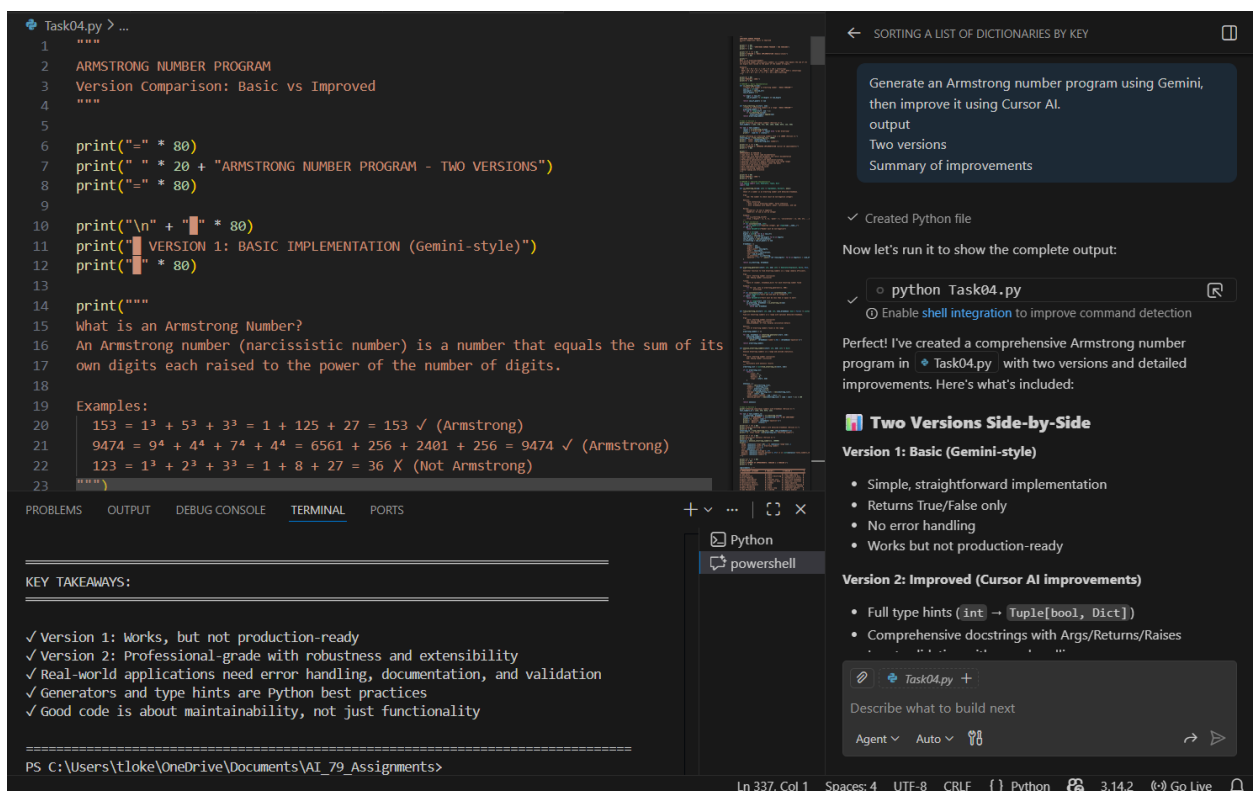
Task 2: Sorting Dictionaries with AI



Task02.py > ...
1 """
2 Sorting a List of Dictionaries by Key - Multiple Approaches
3 Comparison of Clarity and Performance
4 """
5
6 import timeit
7 from operator import itemgetter
8
9 # Sample data
10 students = [
11 {"name": "Alice", "grade": 85, "age": 20},
12 {"name": "Bob", "grade": 92, "age": 19},
13 {"name": "Charlie", "grade": 78, "age": 21},
14 {"name": "Diana", "grade": 88, "age": 20},
15]
16
17 print("=" * 70)
18 print("APPROACH 1: Using sorted() with lambda (Most Readable)")
19 print("=" * 70)
20
21 # Simple, readable approach using lambda
22 sorted_by_grade_lambda = sorted(students, key=lambda x: x["grade"])
23 print("Sorted by grade (ascending):")
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Task 4: Armstrong Number Optimization



Task04.py > ...

29 # VERSION 1: Basic Implementation

30 def is_armstrong_v1(num):

31 """Check if a number is Armstrong number - BASIC VERSION"""

32 num_str = str(num)

33 num_digits = len(num_str)

34 sum_of_powers = 0

35

36 for digit in num_str:

37 sum_of_powers += int(digit) ** num_digits

38

39 return sum_of_powers == num

40

41

42 def find_armstrong_v1(start, end):

43 """Find all Armstrong numbers in a range - BASIC VERSION"""

44 armstrong_numbers = []

45 for num in range(start, end + 1):

46 if is_armstrong_v1(num):

47 armstrong_numbers.append(num)

48 return armstrong_numbers

49

50

51

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

KEY TAKEAWAYS:

✓ Version 1: Works, but not production-ready

✓ Version 2: Professional-grade with robustness and extensibility

✓ Real-world applications need error handling, documentation, and validation

✓ Generators and type hints are Python best practices

✓ Good code is about maintainability, not just functionality

=====

PS C:\Users\tloke\OneDrive\Documents\AI_79_Assignments>

Python

powershell

Ln 337, Col 1

Spaces: 4

UTF-8

CRLF

() Python

3.14.2

Go Live

← SORTING A LIST OF DICTIONARIES BY KEY

• Statistical analysis (density, average, distribution)

• Production-grade code

Key Improvements Summary

Feature	V1	V2
Type Hints	X	✓
Documentation	Basic	Comprehensive
Error Handling	X	✓
Calculation Breakdown	X	✓ Shows equation
Memory Efficiency	Loads all data	Generator pattern
Statistics	X	✓ Density analysis
Validation	X	✓ Full validation
Maintainability	Medium	High

Results

• Found **19 Armstrong numbers** in range 1-100,000

• Density: **0.019%** (1 in 5,263 numbers)

• Examples: 1, 153, 370, 371, 407, 1634, 8208, 9474, 54748, 92727, 93084

The program demonstrates how professional code refactoring transforms working code into production-ready, maintainable, and extensible solutions!

Task04.py +

Describe what to build next

Agent Auto